



CourseRank: A Social System for Course Planning

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ABSTRACT

Special-purpose social sites can offer valuable services to well-defined, closed, communities, e.g., in a university or in a corporation. The purpose of this demo is to show the challenges, special features and potential of a focused social system in action through CourseRank, a course evaluation and planning social system.

Categories and Subject Descriptors

H.3.4 [Information Storage And Retrieval]: Systems and Software—*Information networks*; H.3.5 [Information Storage And Retrieval]: On-line Information Services—*Data sharing*

General Terms

Design, Human Factors, Management

Keywords

social sites, course planning, tag clouds, flexible recommendations

1. MOTIVATION AND OUTLINE

While general-purpose social sites, such as del.icio.us, Flickr and MySpace, address the resource sharing and networking needs of a wider web user community, *focused* social sites (e.g., [4]) offer valuable services to closed, well-defined, communities. In this demo, we show CourseRank in action, an educational and social site where Stanford students can explore course offerings and plan their academic program. CourseRank enables course planning through a number of social activities and interactions, such as sharing and rating, as well as a number of tools that are geared to our domain.

Using CourseRank, students can search for courses of interest, rank the accuracy of each others' comments and get personalized recommendations. They can check if the courses they have taken (or are planning to take) satisfy the requirements for their major. They can also organize their classes into a quarterly schedule or devise a four year plan. Faculty members and university administrators can also participate, providing useful information for students.

Research on social sites has mainly focused on the usage and evolution of these systems (e.g., [1]) and the use of social

knowledge for tasks such as resource recommendations (e.g., [3]). There has been little research on the social information management aspects. In CourseRank, one of the great challenges is interacting with rich data. For this purpose, we have developed new interaction tools, e.g., for flexible recommendations and search refinement, which we will show in the demonstration.

2. SYSTEM OVERVIEW

CourseRank has several important features that distinguish it from many classical social sites but also from other public course evaluation sites and make it a special type of *focused, closed-community, social site* [2]:

- *Hybrid system.* CourseRank provides access to both official Stanford data (e.g., course descriptions, schedules and results of course evaluations conducted by the university), as in a typical database application, as well as to user-contributed information (e.g., course rankings, comments and questions), as in a typical social system.
- *Rich data.* Courses, unlike books or videos, have to be taken in a certain order and in certain quarters. A course is offered by multiple instructors and may use multiple textbooks. Students enroll in courses and get grades. This richness of data introduces many information management challenges, e.g., for building meaningful recommendations, the system should take into consideration how useful a course is for completing a major.
- *Closed Community with Multiple Constituents.* Many social sites are open and have a single type of user. CourseRank is only available to the Stanford community, where there are three very distinct types of users: (a) *Students* (undergraduate and graduate); (b) *Faculty* who may want to check comments on their courses and compare against other courses; and (c) *Staff*, who can enter requirements for academic programs, and who may want to advise a student on his course plan. It is important to offer specialized features for each of these constituencies, to motivate them to use the site and provide resources that can help the other constituencies.

Figure 3 sketches the several components that comprise CourseRank. CourseRank provides tools similar to the ones found at existing social sites, such as for searching for courses (*Keyword Search*) and evaluating courses and other users' opinions (*Eval*) and for asking questions and providing answers (*Q/A*). In addition, CourseRank offers powerful tools geared to our domain, for example, a tool for planning an academic program (*Planner*) that checks for schedule con-

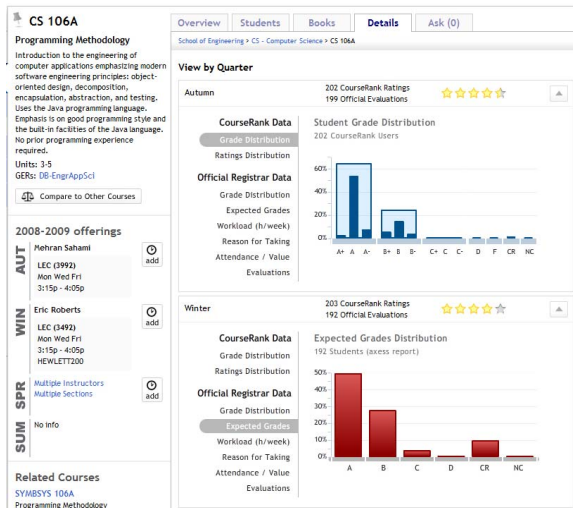


Figure 1: Course description page.

flicts and computes grade point averages, a tool that checks if requirements for a major have been met (*Req Tracker*), a tool for generating statistics for courses (*Statistics*), and a tool for online book exchange (*Book Exchange*). Figure 1 shows a course descriptor page and Figure 2 shows the 4-year course planner.

3. INTERACTING WITH DATA

One of the main challenges we face in CourseRank is interacting with data. Stanford offers a wealth of courses (18,605), but it is hard to navigate through them all using the course catalog. To help students discover courses and track their progress, we have developed a set of tools that are based on new interaction schemes.

Coursecloud. Tag clouds are popular for navigation and visualization purposes over unstructured data because they can highlight the most significant concepts and hidden relationships in the underlying content dynamically. We transfer the idea of tag clouds over structured data and use them to summarize search results over a database.

In particular, the Coursecloud is a tag cloud, where the “tags” are the most representative or significant words found in the results of a keyword search for courses offered by the university. For instance, a student interested in taking a dance class can type the keyword “dance” and get a list of matching courses along with a cloud summarizing course information in this list, as shown in Figure 4. The keyword “dance” is searched in different fields and relations in CourseRank’s database, not just in the main course relation. For example, if there are comments that mention “dance”, the respective courses will appear (in some position) in the results. Searching deeper in the database for the query keywords enables us to discover interesting associations between the results and the search keywords. The results are ranked based on the importance and position of the search keywords associated with them.

The Coursecloud provides concepts related to “dance” that are found in the matching courses, such as “performance”, “Latin America”, and “social dance”. These words may be found in different parts of the database related to the current search. We rank them taking into account both their importance for the particular search and their relative position in the search results. For example, the term “performance” is

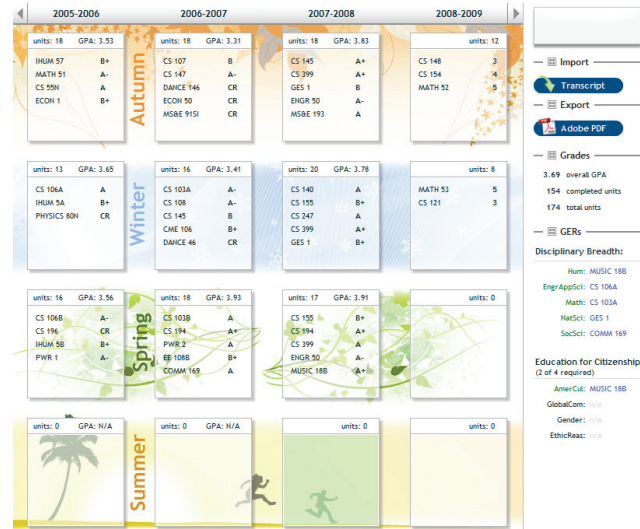


Figure 2: Course planner.

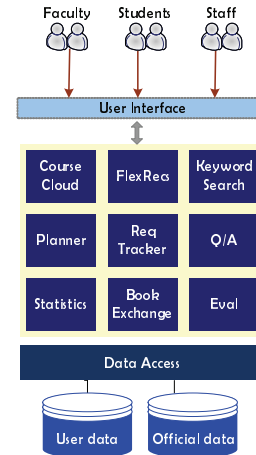


Figure 3: CourseRank System.

found in user comments that refer to “dance” courses with live performances. Terms in the data cloud (as in traditional tag clouds) are hyperlinks. The searcher can click on a term from the data cloud to refine search results. For example, she can click on “Latin America”. The cloud is updated accordingly to reflect the new, refined, results (on “Latin America” and “dance” courses).

Flexible recommendations. Traditional recommendation engines can select popular courses or courses that “people like me” have taken, but they offer little control to the users to personalize their recommendations. For instance, students want to specify the type of course they are interested in (e.g., a biology class, something that satisfies Stanford’s writing requirement), to request recommendations from a peer group they select (e.g., students in their major with similar grades, or freshmen only), and so forth.

CourseRank has a novel recommendation engine that allows flexible recommendations to be easily defined, customized, and processed. A given recommendation approach can be expressed declaratively as a high-level workflow over structured data and then executed by the underlying engine. At the heart of the engine lies a special recommend operator, which takes as input a set of tuples and ranks them by comparing them to another set of tuples. The operator may call

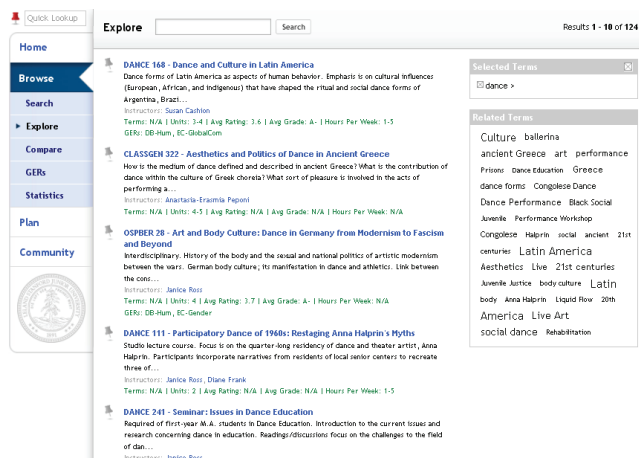


Figure 4: Searching for “dance”.

upon functions in a library that implement common tasks for recommendations, such as computing the Jaccard or Pearson similarity of two sets of objects, and it may be combined with other recommend operators and traditional relational operators such as select and join.

A designer can easily create multiple, customizable workflows for different recommendation types. The end user can select from them and enter parameters for workflows in order to get more accurate and personalized recommendations. This selection is done through a GUI. For instance, the user may specify that her recommendations be based on the courses taken by students in her class. This choice gets translated into a select condition, which is passed to a set of SQL statements and executed transparently to the user.

Course requirements. Program requirements can be very complex. Capturing and understanding such knowledge can enable the system help students track their progress. Commercial systems, such as Peoplesoft can represent this knowledge but it is hard to enter and update data.

For this purpose, we have developed a simple, formal model that allows CourseRank to represent requirements. For example, the math requirement specifies that in order to collect 23 units on math, one has to take `math41` and `math42` or `math19`, `math20` and `math21` and two of `math51`, `math103`, `CME100` and `cs156` not counting some combinations. This requirement can be represented in our model as follows:

```
Math ( 23 units )
  [ ((Math41, Math42) or (Math19, Math20, Math21)),
    ( two of (Math 51, Math103, CS156,...)
  not counting (Math51, Math103), (Math51, CME100, ...)) ]
```

CourseRank has a simple interface that allows program administrators to enter requirements. For a student, it can check whether the requirements for a major are met based on the set of courses taken. Figure 5 shows the page of the requirements tracker. We have found that the requirements checking problem is NP-hard. One reason is that a course can be used in satisfying many requirements but it counts only once. Our checker is fast but has some false negatives.

4. DEMONSTRATION

We will demonstrate the several tools that comprise CourseRank. We will browse the course pages of different courses in Stanford, and show official information, such as course de-

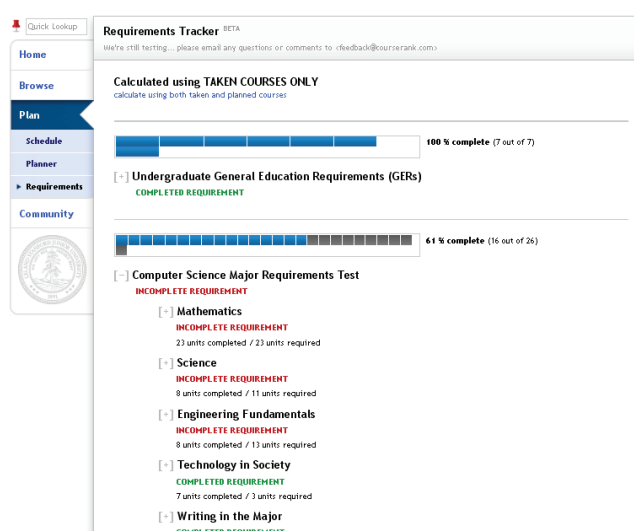


Figure 5: Requirement tracker page.

scription and course evaluations, and unofficial information, such as questions and answers, comments, and statistics. We will navigate through the Book exchange (wiki) pages, and see how the students can report and find books for courses. The Book component pulls additional information for a book reported by a student from Amazon. We will see how the Course planner helps students organize their four-year plan and checks for schedule conflicts and computes grade point averages. We will also show how CourseRank advises students on which requirements have been met and allows trying different scenarios for meeting requirements for a major.

We will demonstrate the Coursecloud. We will show how we can refine our searches by adding or removing terms in a search and navigate using the cloud. To illustrate how the Coursecloud helps find diverse and serendipitous course offerings, we will demonstrate different search scenarios, such as finding a course for English or learning about civil rights, and we will compare with the results we can get using more traditional search/browse interfaces.

Finally, we will explore flexible recommendations and show how students can customize recommendations provided by the system through an advice panel. Changing the various options that the advice panel offers, a student is able to control the recommendation process and experiment with different types of course recommendations. For example, she can specify that the recommendations be based on her course history or based on what students in her major have taken, or students with similar grades or tastes, and so forth.

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